

Setup domain Controller



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Introduction

Active Directory (AD) is a centralized directory service provided by Windows Server that enables organizations to manage identities, resources, and security policies across a network. It acts as the backbone of enterprise IT infrastructure by storing information about users, computers, groups, and services in a structured database, and by authenticating and authorizing access using protocols such as Kerberos and LDAP. In modern environments, Active Directory ensures that administrators can enforce consistent security controls, streamline user management, and maintain operational efficiency. The Active Directory Administrative Center (ADAC) further enhances this capability by offering a modern interface integrated with PowerShell, enabling both graphical

Main Content

Active Directory is built on a hierarchical and logical structure consisting of forests, domains, organizational units (OUs), and objects. A **forest** represents the highest level and acts as a security boundary, while a **domain** serves as a container for users, computers, and policies.

Promote to Domain Controller

1. Look for “Flag with Warning” on the Active Directory Domain Services page.
2. Click on “Promote this server to a domain controller”.

Forest and domains are separate concepts in AD:

- **Forest:** A unit of network services, composed of domains that trust one another. For example, having labs.local as the forest name might mean your computers within these forests trust each other.
- **Domain:** A component of a forest. All users or computers must be part of at least one domain.

So when you promote a server to a domain controller, it joins that specific *domain*, not necessarily the entire *forest*.

Setup

What We Are Building (Quick Context)

You are setting up:

- 1 × Windows Server (DC01) → runs AD DS + DNS
- 1 × Windows 10/11 (CLIENT01) → joins the domain
- Domain Name → lab.local

This simulates a real enterprise identity infrastructure using Windows Server.

Lab Requirements

Machines

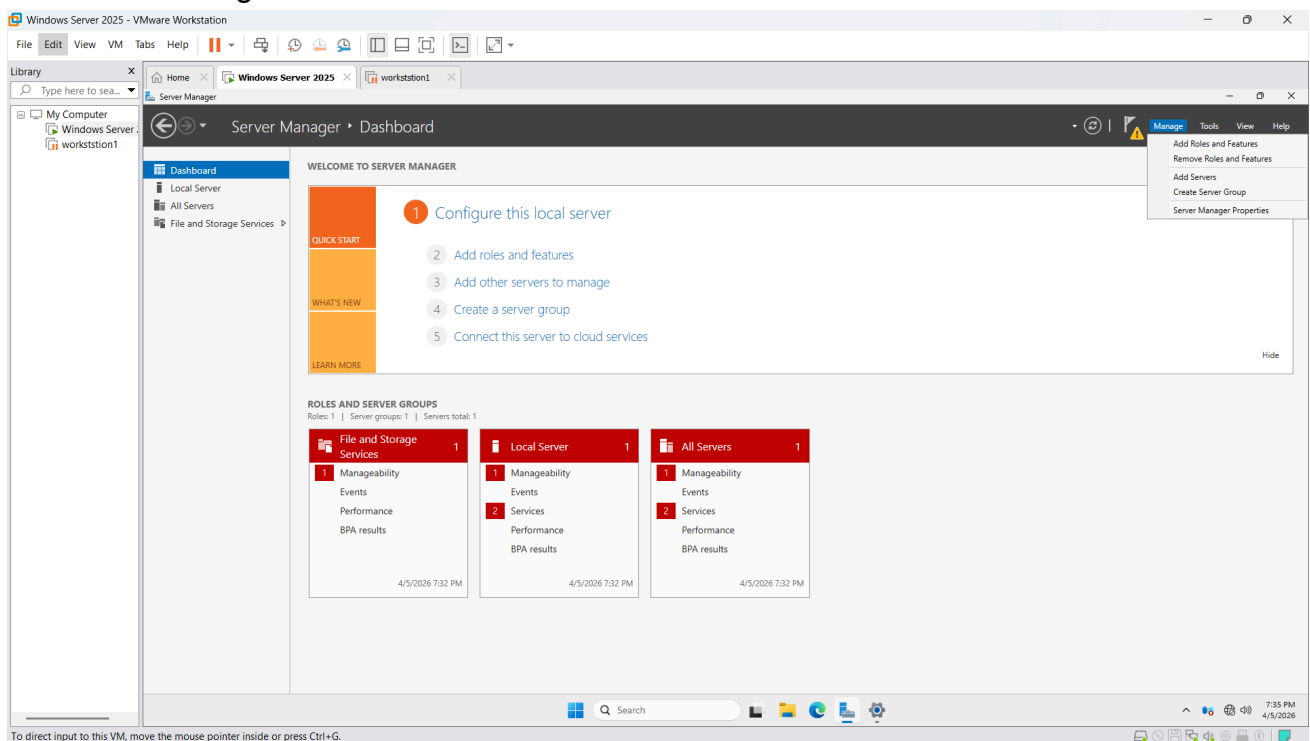
- 1 × Windows Server (DC01)
- 1 × Windows 10/11 (CLIENT01)

Network

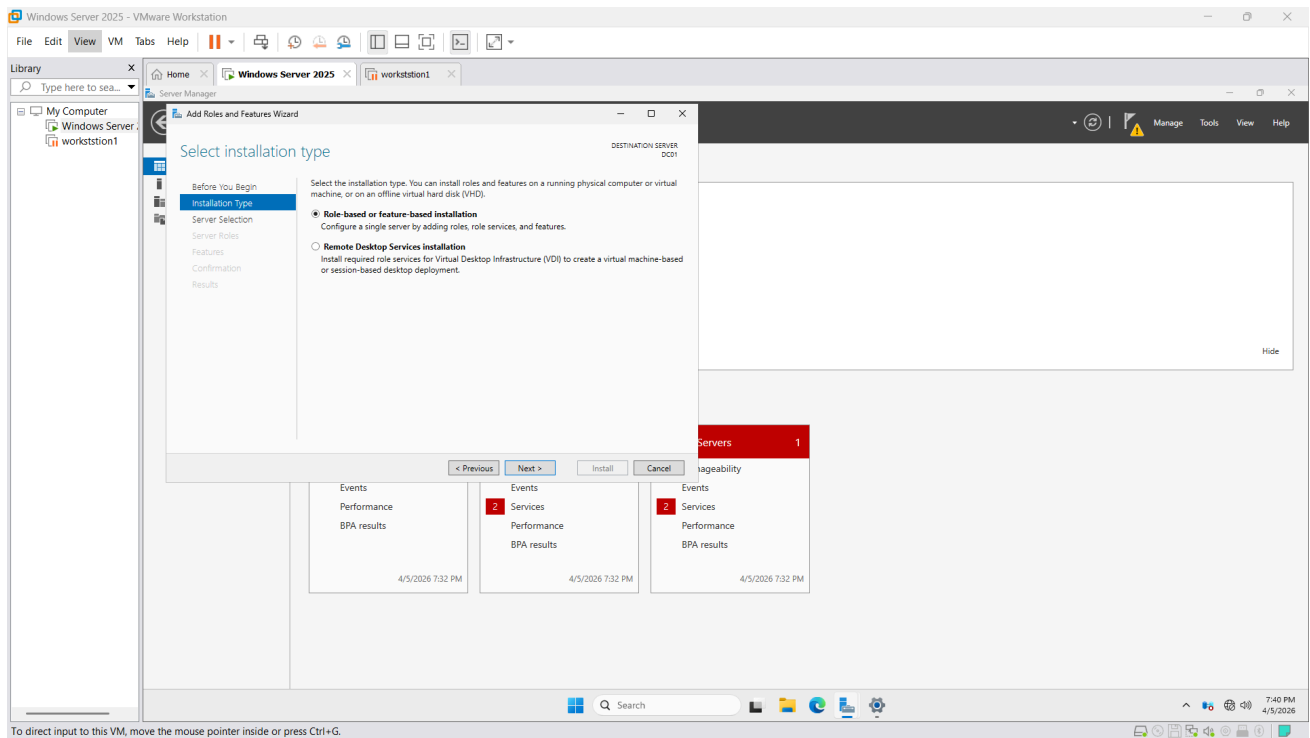
- Same VM network (NAT or Host-only)

*** Open Service Manager

- Step 1
 - ** click on manage > add roles and features

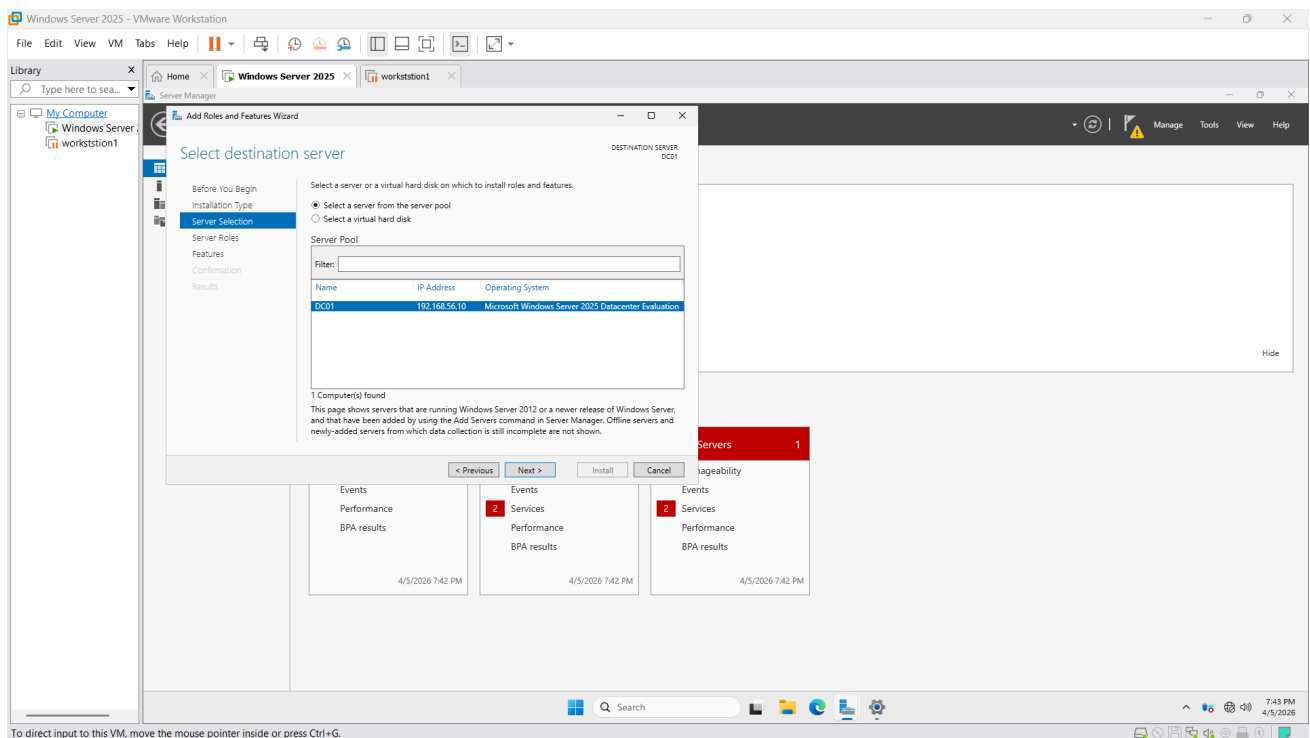


- Step 2
 - ** click on next > select Role-based or feature-base installation then next.



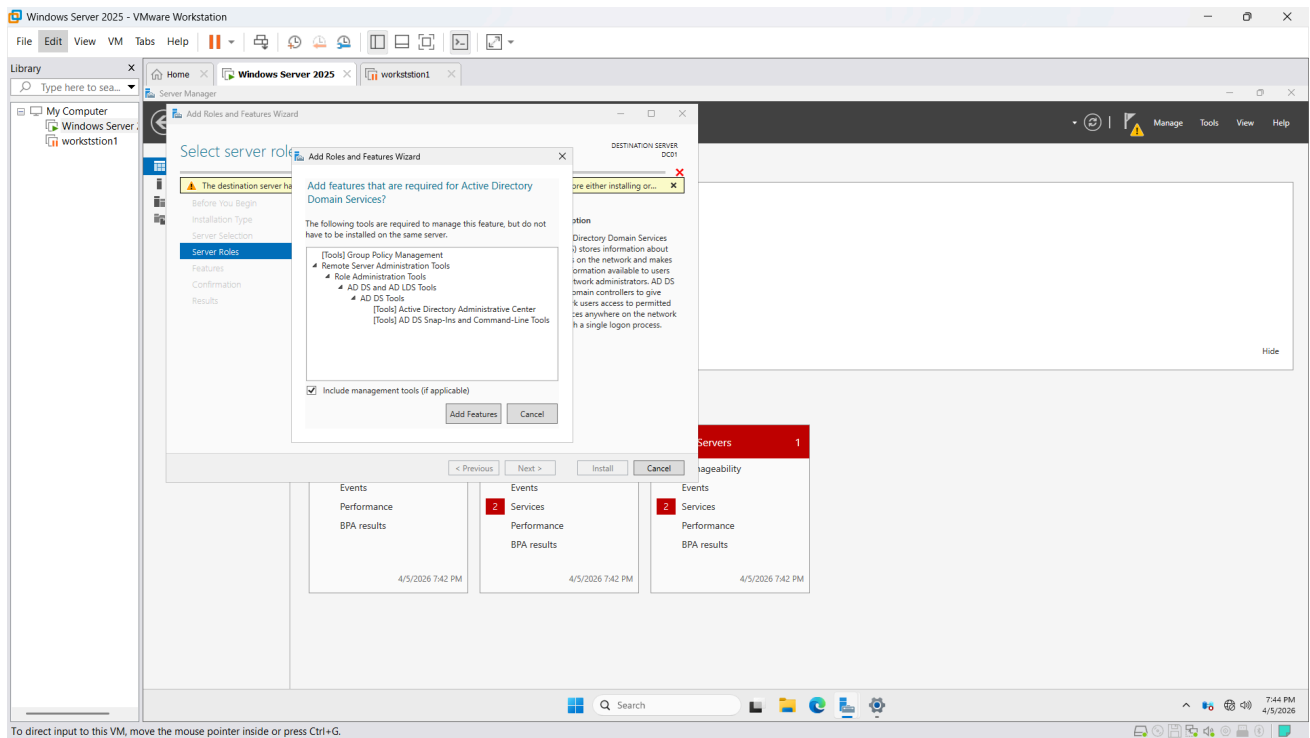
- Step 3

**** Select your Server name mine is DC01 if you didnot create, Create one it is just your pc name. Then next**



- Step 4

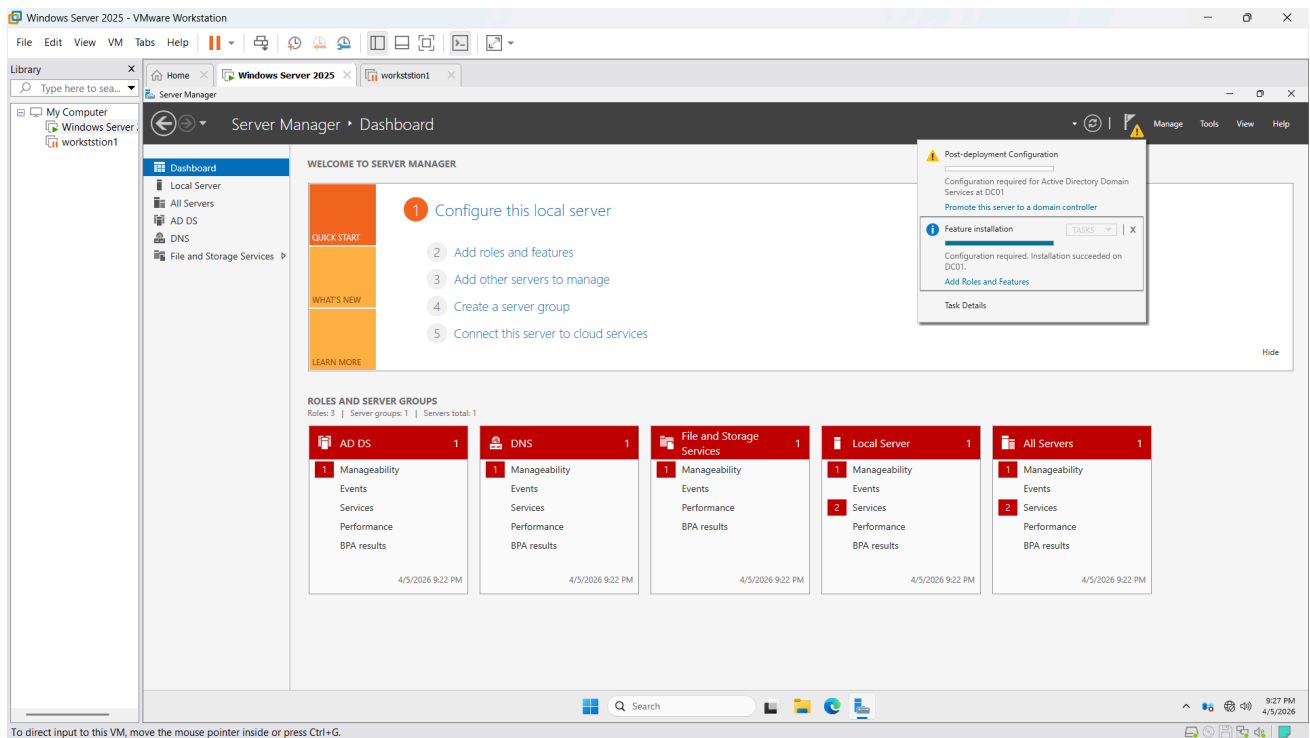
**** Select Active Directory Domain Services and DNS Server then click on add features.**



Promote to Domain Controller

- Step 1

Look for “Flag with Warning” on the Active Directory Domain Services page, click on “Promote this server to a domain controller”.

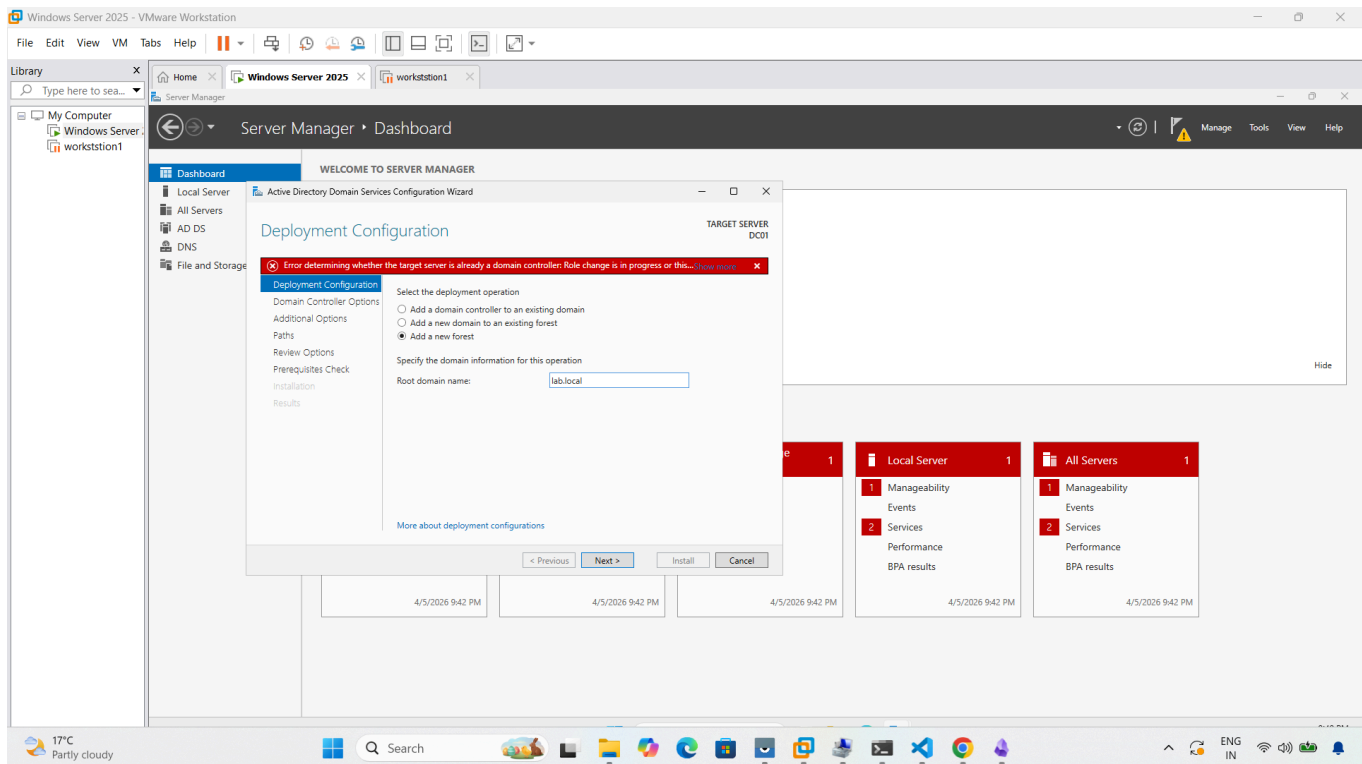


Step 2

Click on the **New Forest** option: Select your Forest Name and then proceed with the forest creation steps

A **Forest** is a unit of network services composed of domains that trust one another. For

example, having labs.local as the forest name means all computers within this forest trust each other.



Step 3

-This is most important part of this whole setup always use latest window server like for now window server 2025 for security purpose, Then check DNS server and GC.

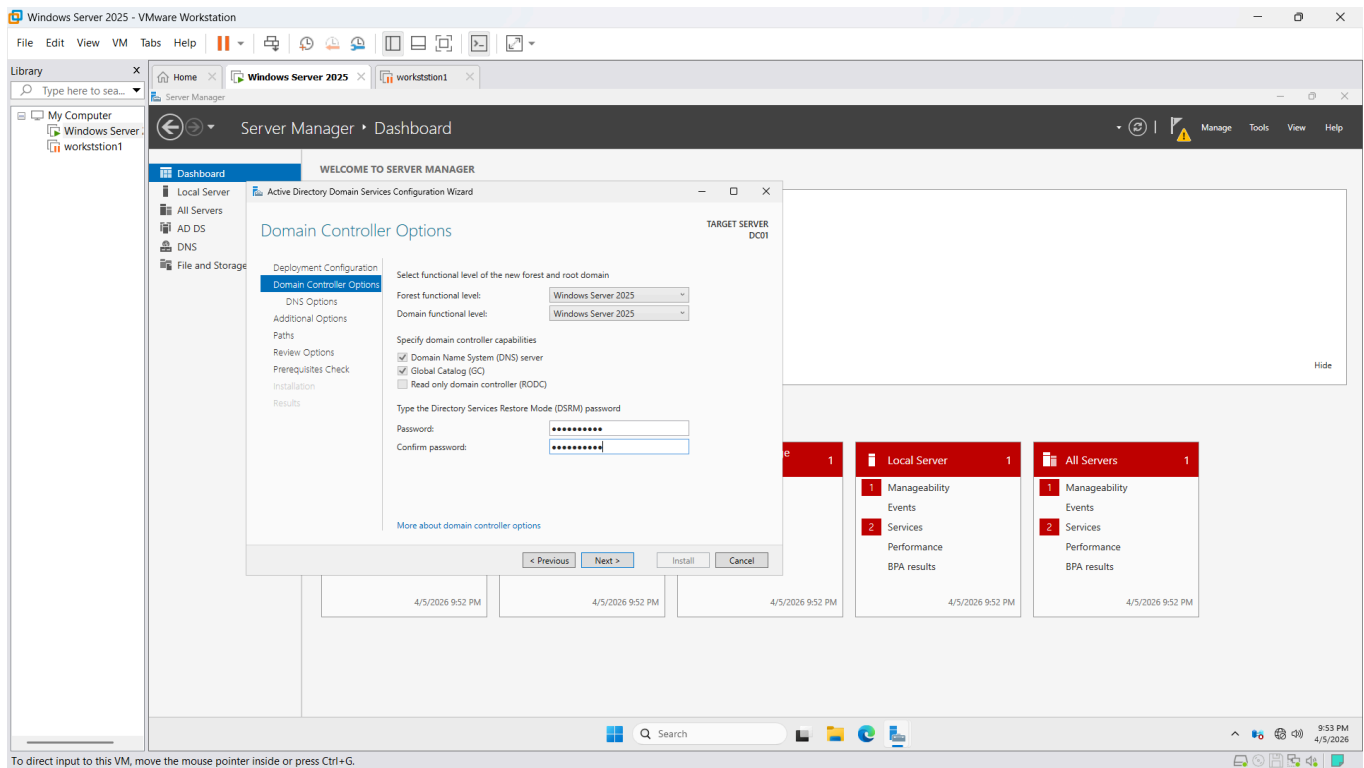
- Why this matter?

-DNS - Required for AD

-GC - Enable fast authentication and search.

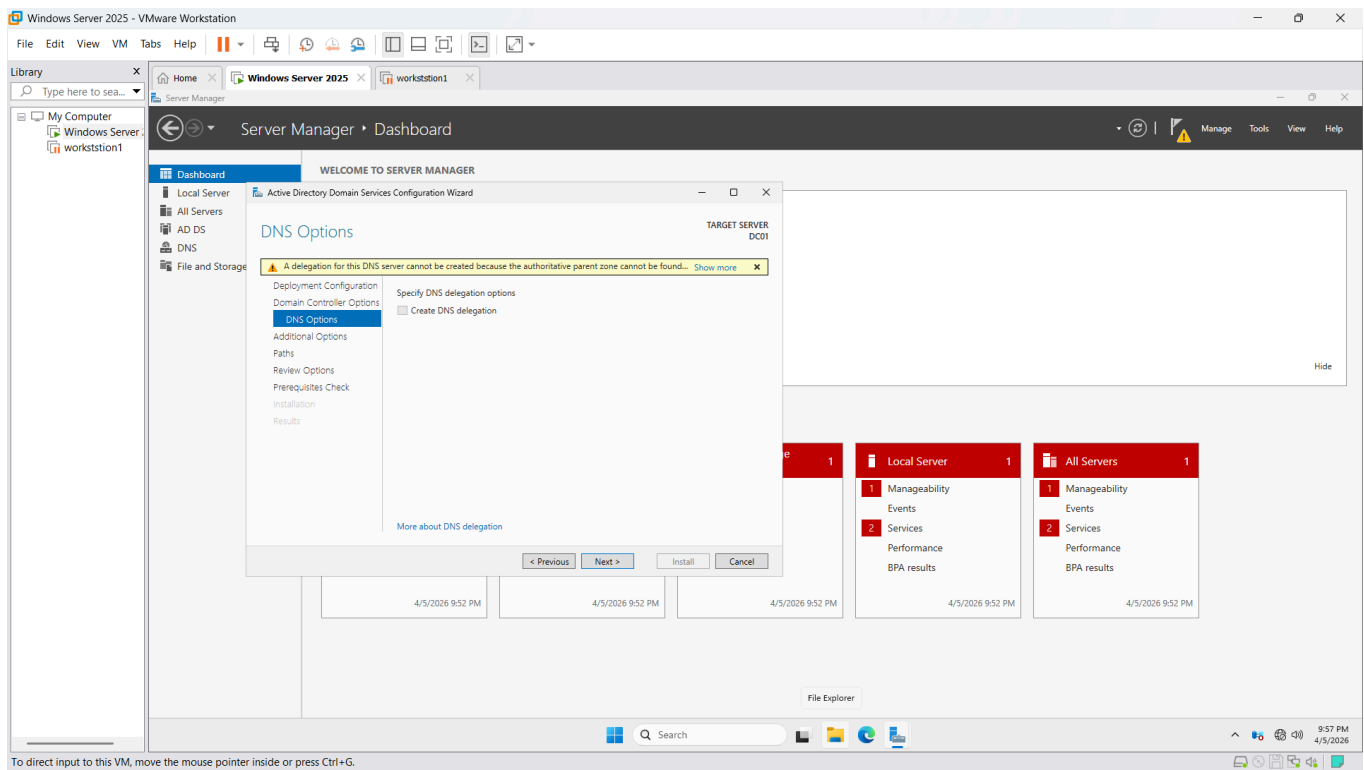
- leave uncheck rest

- Set Good Strong password



Step 4

- For now we donot need DNS Delegation as it is for main/sub dns for big org.



Step 5

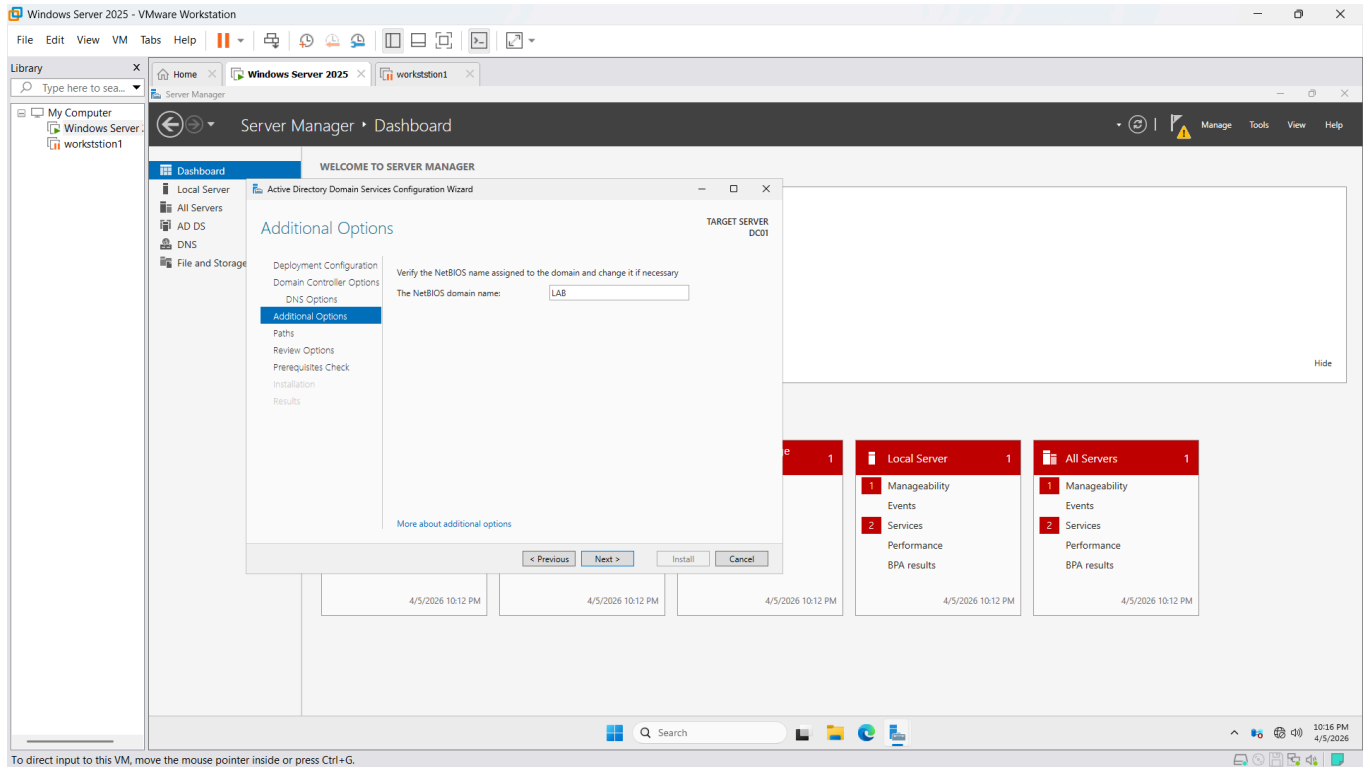
- Do not Change NetBIOS Domain Name = LAB

This is the short name of your domain: `lab.local`

NetBIOS name - LAB

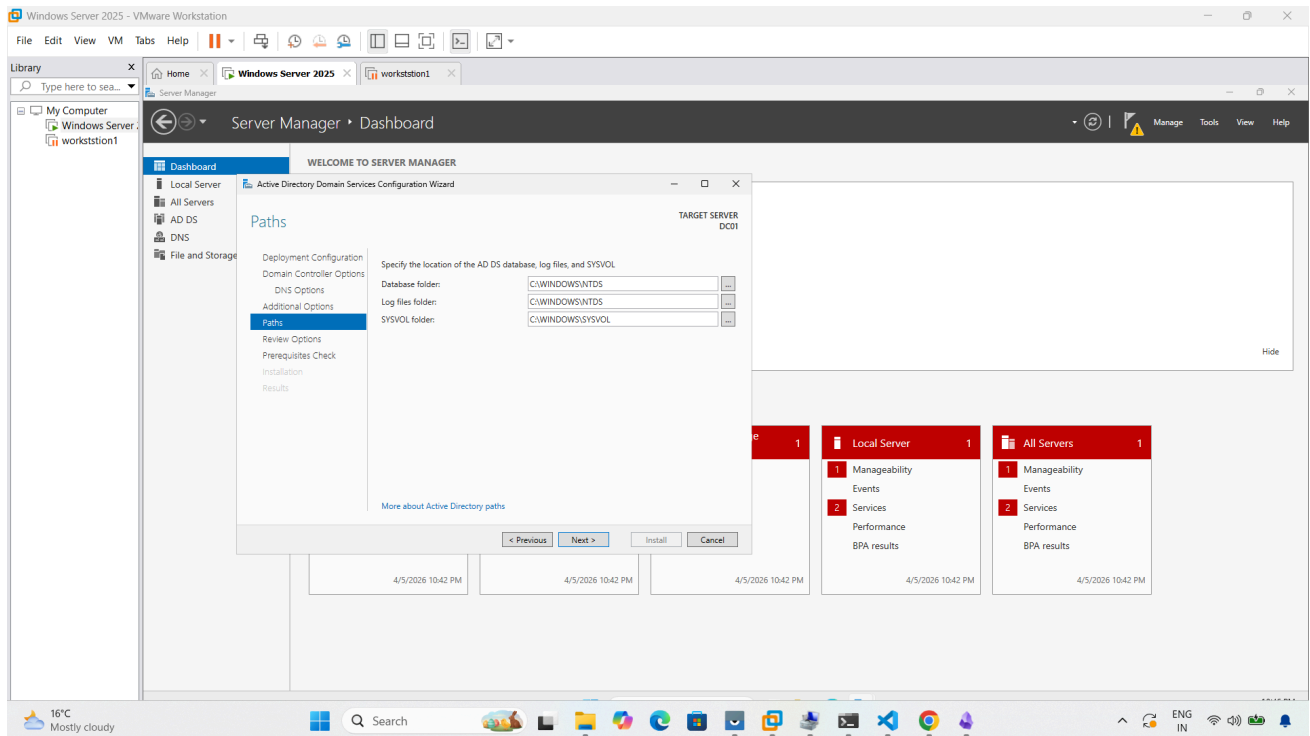
Why NetBIOS Exists?, Before modern DNS, Windows used NetBIOS names for: Login , Network, discovery, Legacy systems. Even today, it's still used in:

`LAB\Administrator`



Step 6:

- Donot change anything next.



now install after pre-requisites check , make sure your Administrator password is strong to pass this check and restart.

